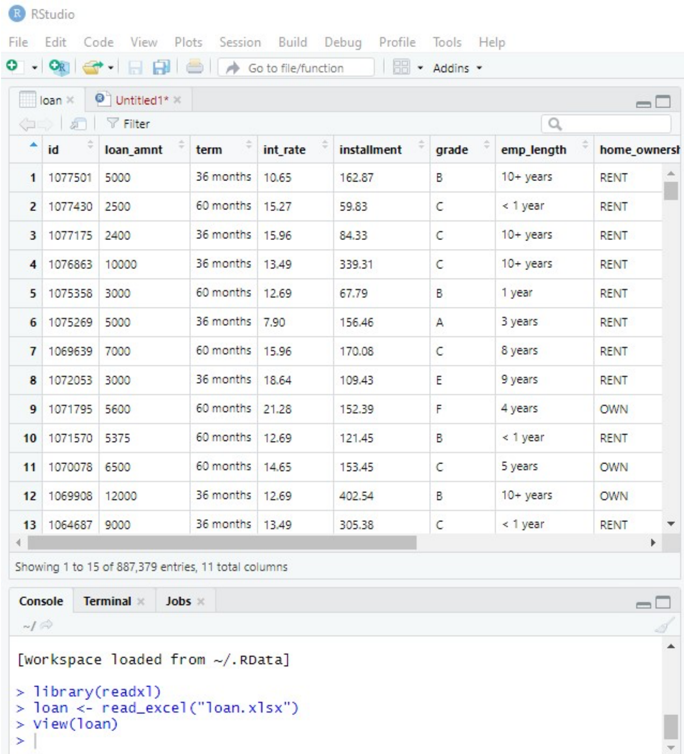
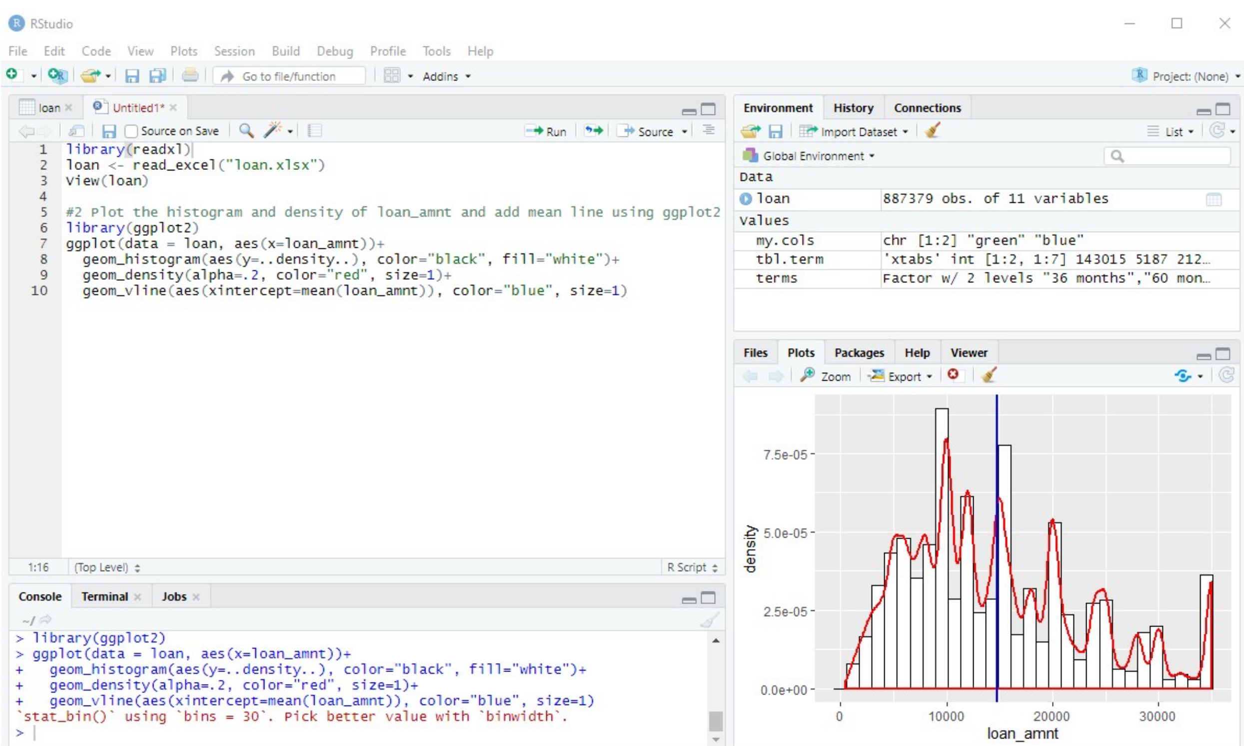
Week 6 Sean Graham

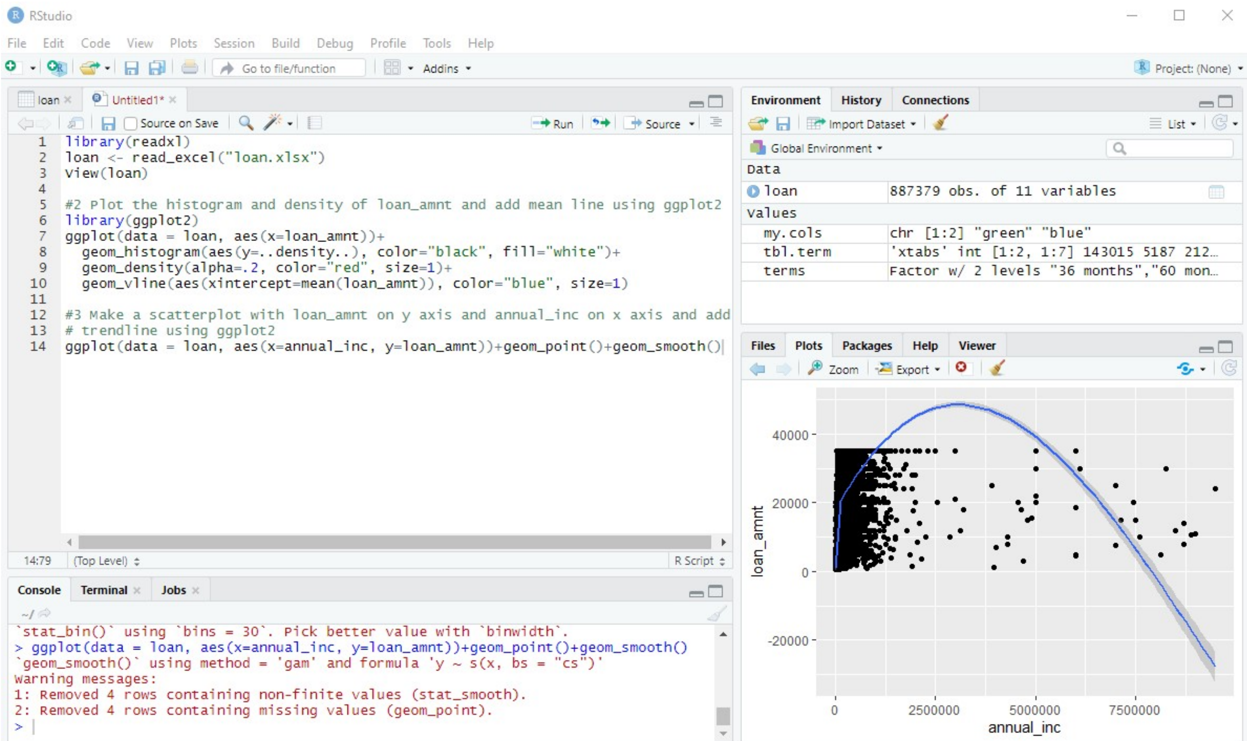
1.



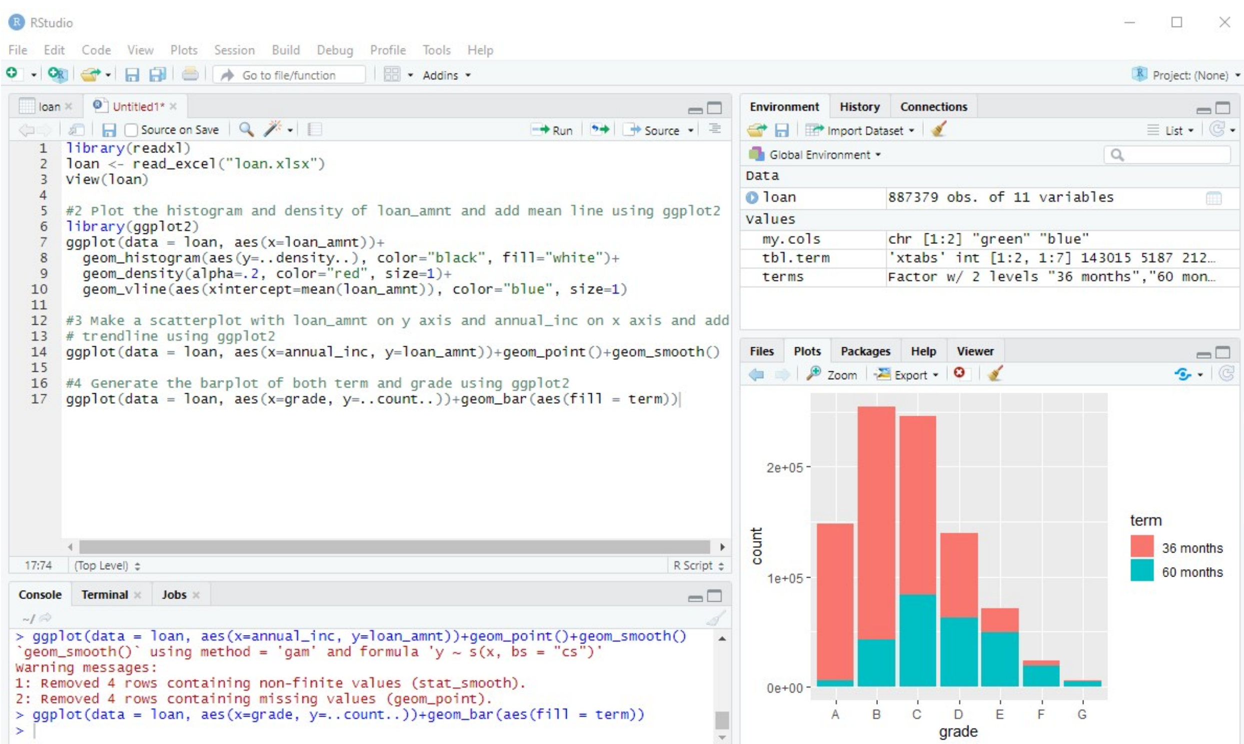
2.



3.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
library(readxl)
loan <- read_excel("loan.xlsx")
View(loan)
#2 Plot the histogram and density of loan_amnt and add mean line using ggplot2
library(ggplot2)
ggplot(data = loan, aes(x=loan_amnt))+
geom_histogram(aes(y=..density..), color="black", fill="white")+
geom_density(alpha=.2, color="red", size=1)+
geom_vline(aes(xintercept=mean(loan_amnt)), color="blue", size=1)
#3 Make a scatterplot with loan_amnt on y axis and annual_inc on x axis and add
# trendline using ggplot2
ggplot(data = loan, aes(x=annual_inc, y=loan_amnt))+geom_point()+geom_smooth()
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
> ggplot(data = loan, aes(x=annual_inc, y=loan_amnt))+geom_point()+geom_smooth()
`geom_smooth()` using method = 'gam' and formula 'y ~ s(x, bs = "cs")'
Warning messages:
1: Removed 4 rows containing non-finite values (stat_smooth).
2: Removed 4 rows containing missing values (geom_point).
Environment History Connections
Global Environment
Data
loan 887379 obs. of 11 variables
Values
my.cols chr [1:2] "green" "blue"
tbl.term 'xtabs' int [1:2, 1:7] 143015 5187 212…
terms Factor w/ 2 levels "36 months","60 mon…
Files Plots Packages Help Viewer
loan_amnt
40000
20000
0
-20000
0
2500000
5000000
7500000
annual_inc

4.


RStudio
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#3 Make a scatterplot with loan_amnt on y axis and annual_inc on x axis and add
# trendline using ggplot2
ggplot(data = loan, aes(x=annual_inc, y=loan_amnt))+geom_point()+geom_smooth()
#4 Generate the barplot of both term and grade using ggplot2
ggplot(data = loan, aes(x=grade, y=..count..))+geom_bar(aes(fill = term))
> ggplot(data = loan, aes(x=annual_inc, y=loan_amnt))+geom_point()+geom_smooth()
`geom_smooth()` using method = 'gam' and formula 'y ~ s(x, bs = "cs")'
Warning messages:
1: Removed 4 rows containing non-finite values (stat_smooth).
2: Removed 4 rows containing missing values (geom_point).
> ggplot(data = loan, aes(x=grade, y=..count..))+geom_bar(aes(fill = term))
Environment History Connections
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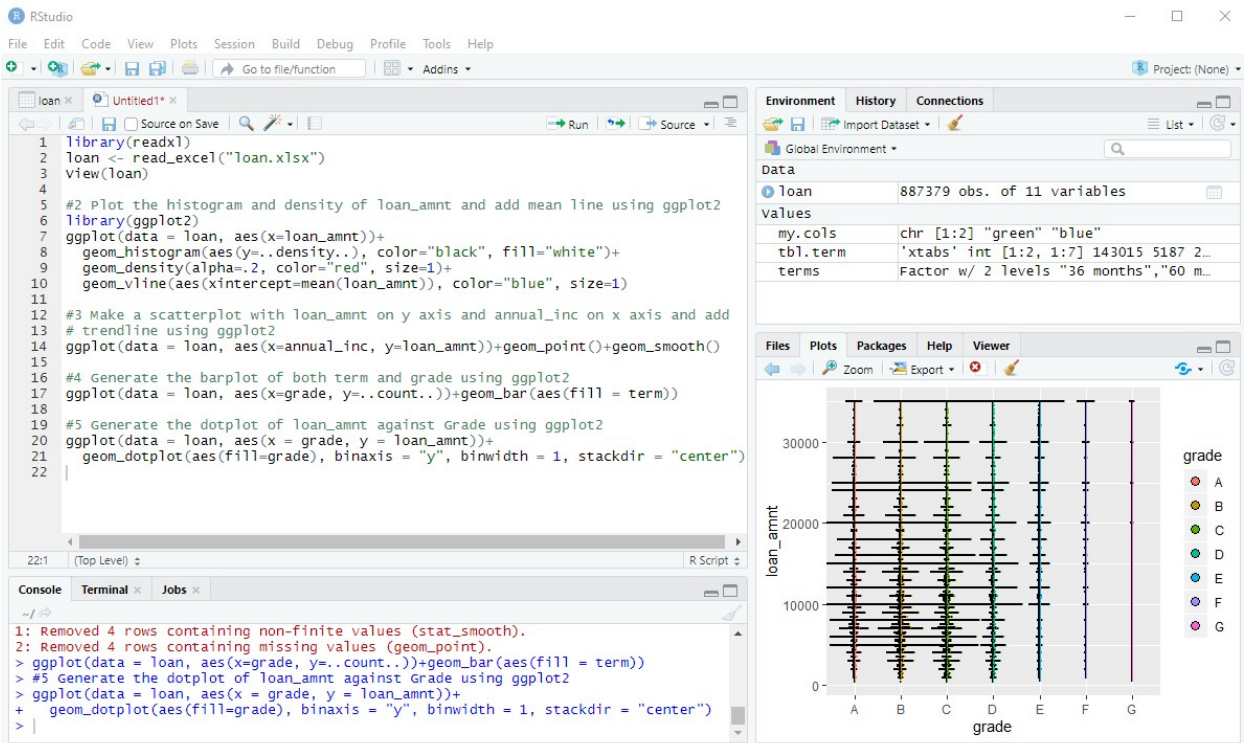
Values
my.cols chr [1:2] "green" "blue"
tbl.term 'xtabs' int [1:2, 1:7] 143015 5187 212…
terms Factor w/ 2 levels "36 months","60 mon…
Files Plots Packages Help Viewer
count
2e+05
1e+05
0e+00
A B C D E F G
grade
term
36 months
60 months

5.



6. Loans with terms of 60 months tend to be larger than those with terms of 36 months. The notches are very narrow, but they're still there, and they don't overlap, so we can say with 95% confidence that the medians are different.